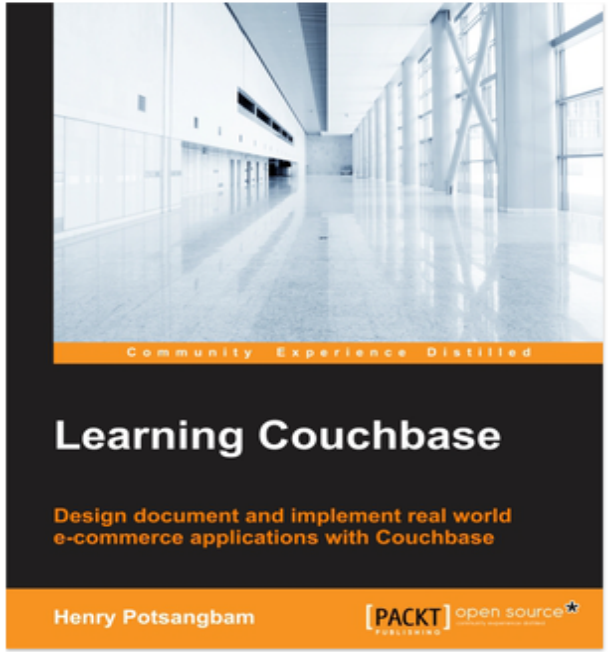
Kafka – Admins & Operations

Henry R.P

- Certified Cassandra Admin
- Mapr Certified – Hadoop Administrator
- IBM Certified Application Developer
- IBM Certified Solution Designer
- SAP Certified ABAP & Portal Consultant .
- CIPM – Certificate in Project Management.
- TOGAF – Enterprise Architect

IT Architect & Corporate trainer
20 +Year of IT Experience

NOSQL, Streaming Platform & Bigdata



Author

Clientele

Tos

Outline

Tos

Introduction
Real-Time Processing is Becoming Prevalent
Kafka: A Stream Data Platform
Kafka Fundamentals
An Overview of Kafka
Kafka Producers

Kafka CLI
Zookeeper

Managing a Kafka Cluster
Installing and Running Kafka
Monitoring Kafka
Basic Cluster Management
Log Retention and Compaction

Providing Durability
Basic Replication Concepts
Durability Through Intra-Cluster Replication
Writing Data to Kafka Reliably
Broker Shutdown and Failures
Controllers in the Cluster
The Kafka Log Files
Offset Management

Producer Architecture

Java API - Kafka Consumer.
Failover and Consumer Failover

Consumer
Push vs. pull
Message Delivery
Reprocess of the Failed Message
Message Offset management .

Designing for High Availability
Kafka Reference Architecture
Brokers

Optimizing Kafka Performance
Producer Performance
Broker Performance
Load Balancing Consumption
Consumption Performance
Performance Testing

Kafka Security
SSL for Encryption and Authentication
SASL for Authentication
Data at Rest Encryption

Kafka – DR – (Kafka mirroring (MirrorMaker))

Integrating Systems with Kafka Connect
The Motivation for Kafka Connect
Types of Connectors
Kafka Connect Implementation

Monitoring Kafka

Introduce Yourself.

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Name
Year of Experience.
Skills Level
    Java / Linux
    Messaging System / Kafka
Expectation, if any.

Note: Basic knowledge of Java & Linux are required.

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Schedule

Time	
9.30 – 11.00 AM	Session I
11.00 AM to 11.15 AM	Tea Break
11.15 AM to 12.45 PM	Session II
12.45 PM to 1.45 PM	Lunch Break
1.45 PM to 3.15 PM	Session III
3.15 PM to 3.30 PM	Tea Break
3.30 PM to 5.30 PM	Session IV

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Kafka – An Overview

Kafka growth exploding

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- 1/3 of all Fortune 500 companies
- Top ten travel companies, 7 of ten top banks, 8 of ten top insurance companies, 9 of ten top telecom companies
- LinkedIn, Microsoft and Netflix process 4 comma message a day with Kafka (1,000,000,000,000)
- Real-time streams of data, used to collect big data or to do real time analysis (or both)

Why Kafka is Needed?

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- Real time streaming data processed for real time analytics
- Service calls, track every call, IOT sensors
- Apache Kafka is a fast, scalable, durable, and fault- tolerant publish-subscribe messaging system
- Kafka is often used instead of JMS, RabbitMQ and AMQP
- higher throughput, reliability and replication

Kafka Use Cases

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- Stream Processing
- Website Activity Tracking
- Metrics Collection and Monitoring
- Log Aggregation
- Real time analytics
- Capture and ingest data into Spark / Hadoop
- CRQS, replay, error recovery
- Guaranteed distributed commit log for in-memory computing

Kafka: A Stream Data Platform

Kafka Streaming Architecture

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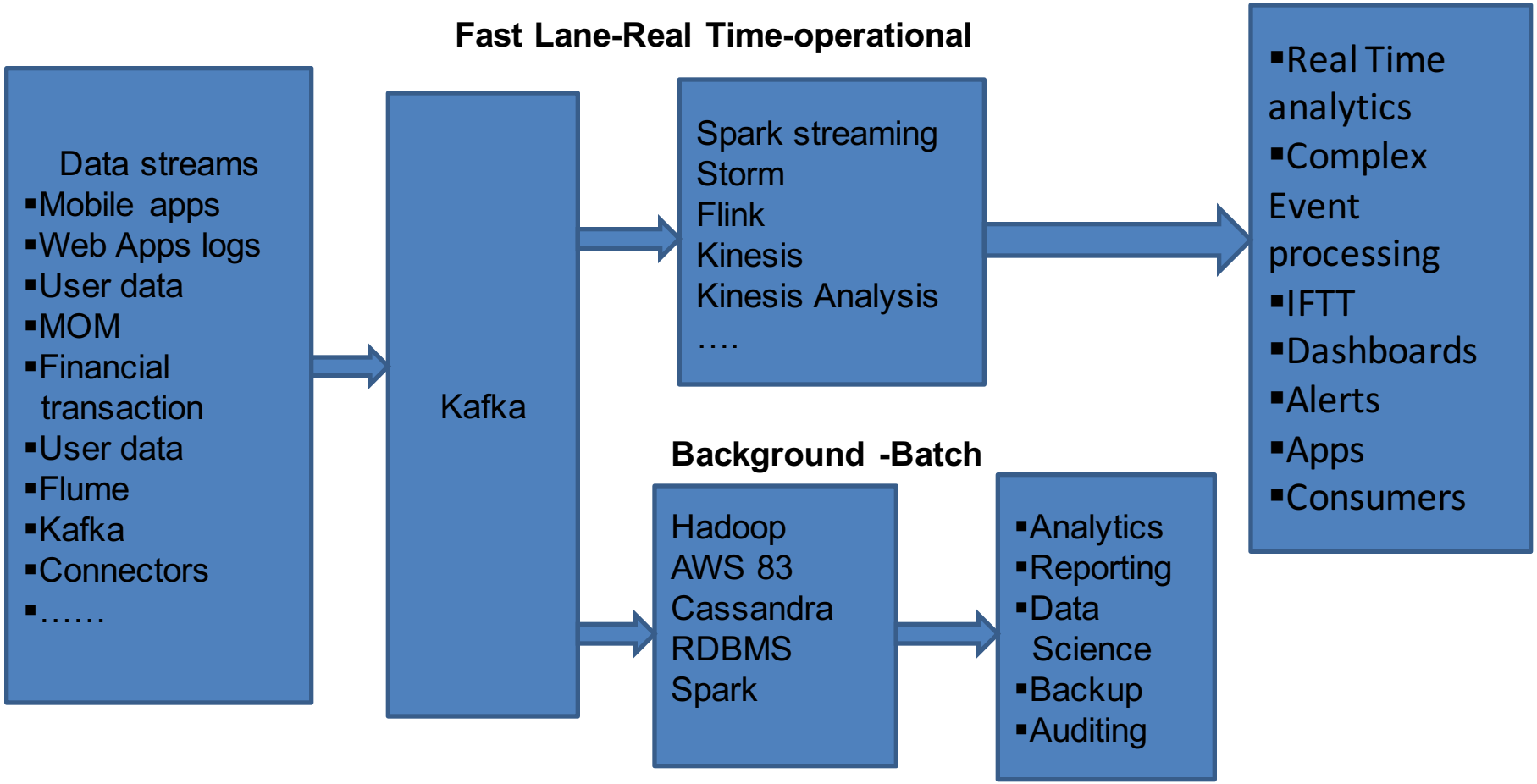

What is Kafka?

What is Kafka?

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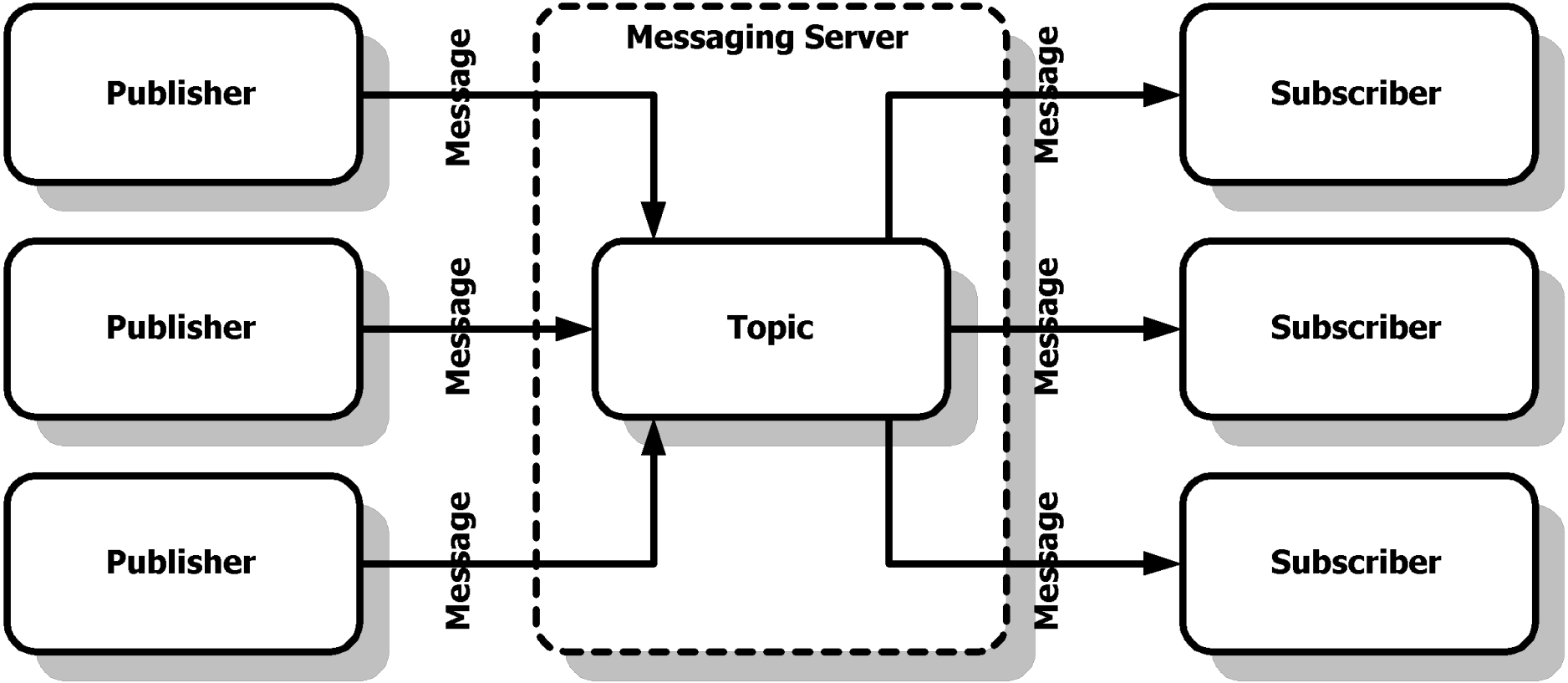

Kafka Decoupling Data Streams

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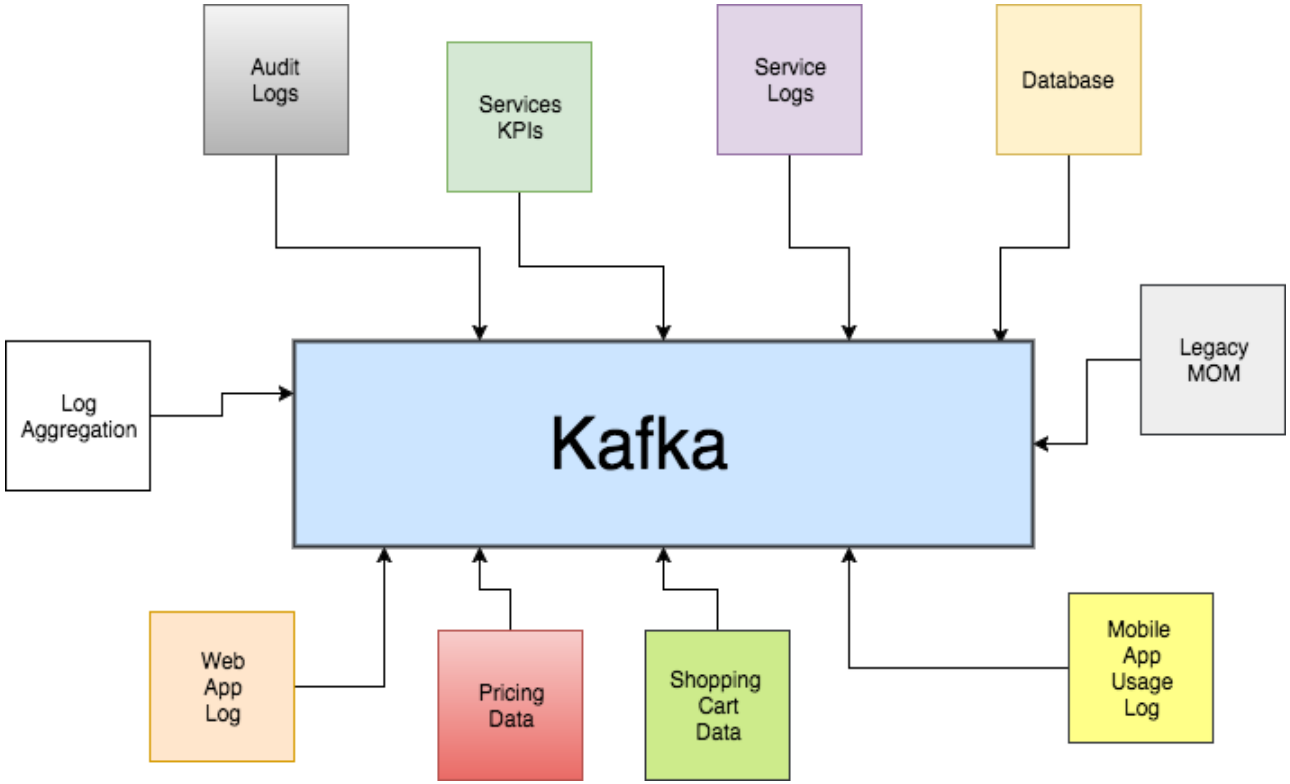

Don't couple the stream

Installation steps

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Installing Java

#tar -xvf jdk-8u45-linux-x64.tar.gz -C /opt

Set in the path variable and JAVA_HOME

Include in the profile a follow

```
[root@tos opt]# more ~/.bashrc
# .bashrc

# User specific aliases and functions

alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'

export JAVA_HOME=/opt/jdk1.8.0_45

export PATH=$PATH:$JAVA_HOME/bin
# Source global definitions
if [ -f /etc/bashrc ]; then
        . /etc/bashrc
fi
[root@tos opt]#
```

Installation steps…

Tos

Installing Zookeeper

The following example installs Zookeeper with a basic configuration in `/opt/zookeeper`, storing its data in `/opt/data/zookeeper`:

```
# tar -xvf zookeeper-3.4.12.tar.gz -C /opt

# mv zookeeper-3.4.12 /opt/zookeeper
# mkdir -p /opt/data/zookeeper
# vi /opt/zookeeper/conf/zoo.cfg
  tickTime=2000
  dataDir=/opt/data/zookeeper
  clientPort=2181
# /opt/zookeeper/bin/zkServer.sh start
```

```
[root@tos opt]# /opt/zookeeper/bin/zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
[root@tos opt]#
```

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Installing a Kafka Broker

The following example installs Kafka in `/opt/kafka`, configured to use the

Zookeeper server started previously and to store the message log segments stored in /tmp/kafka-logs:

```
# tar -zxf kafka_2.12-1.1.0.tgz –C /opt
# mv kafka_2.12-1.1.0 /opt/kafka
# mkdir /opt/data/kafka-logs
# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/server.properties
```

```
[root@tos opt]# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/s
erver.properties
[root@tos opt]# jps
3476 Kafka
3499 Jps
2895 QuorumPeerMain
[root@tos opt]#
```

Lab : **Installation of Kafka**